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Analyzing the role of consumer behavior in coping with intermittent supply in water distribution systems

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ABSTRACT

A substantial number of water distribution systems (WDS) worldwide are operated as intermittent water supply (IWS) systems, delivering water to consumers in irregular and unreliable manners. The IWS consumers commonly adapt to flexible consumption behaviors characterized by storing the limited water available during shorter supply periods in intermediate storage facilities for subsequent usage during more extended nonsupply periods. Nevertheless, the impacts of such consumer behavior on the performance of IWS systems have not been adequately addressed. Toward this direction, this article presents a novel open-source Python-based simulation tool (EPyT-IWS) for WDS, virtually acting like an IWS modeling extension of EPANET 2.2. The applicability of EPyT-IWS was demonstrated by conducting hydraulic simulations of a typical WDS with representative IWS attributes. Different IWS operation cases were considered by varying the amount and consistency of the water availability to the consumers. EPyT-IWS outputs showed that domestic storage of water within underground tanks and subsequent pumping into overhead tanks allows consumers to cope with the intermittent water availability and suitably meet their demands. Besides the interval, the clock time of the water supply was predicted to influence IWS consumers' ability to meet water demands.

Key words: EPANET-Python, intermittent water supply, pressure-dependent analysis, water distribution, water scarcity, water security

HIGHLIGHTS

- Intermittent water supply (IWS) consumers adapt to flexible consumption behaviors.
- Store the water available during shorter supply periods for subsequent usage.
- EPyT-IWS can simulate this typical consumer behavior more realistically.
- Location of consumers controls water withdrawal and storage during supply hours.
- Clock time of supply influences consumers' ability to meet water demands.

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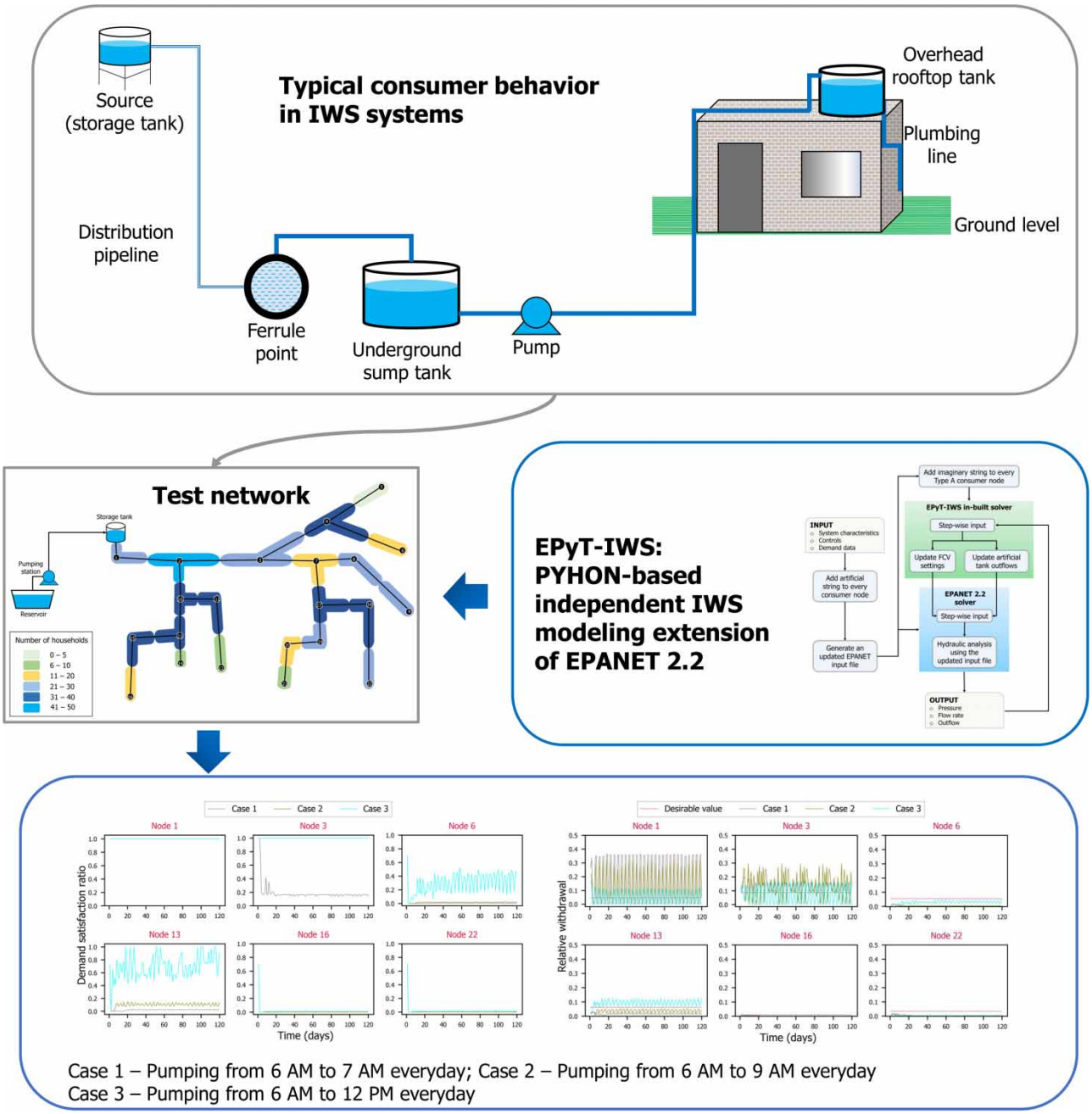
GRAPHICAL ABSTRACT



INTRODUCTION

Water distribution systems (WDS) are designed to deliver consumers adequate drinking water, meeting necessary quality criteria. Although WDS of the developing regions of the globe is designed initially to provide water continuously per specified standards, most operate inconsistently. Specifically, in South Asian countries, particularly India, most WDS provide water to consumers for less than 24 h a day (Charalambous & Laspidou 2017; Ghorpade *et al.* 2021). These systems are termed intermittent water supply (IWS) systems. Kumpel & Nelson (2016) estimated that an alarming (~370 million) population in India relies on IWS systems to meet their water demands.

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Several causes, such as population growth, urbanization, increased occurrence of extreme weather events, inadequate accessibility of freshwater, inadequate sizing of distribution network, and water losses, can be attributed to the IWS operation of WDS (Loubser *et al.* 2020). Ghorpade *et al.* (2021) reasoned that once any WDS falls into the 'vicious' trap of IWS operation, it might become challenging to climb up because of the intricate relationships between several drivers (design, operation, maintenance, institutional capacity, consumer satisfaction, revenue generation, etc.) that govern its performance.

Altogether, IWS systems can be divided into three categories – predictable (nonsupply occurs within a predictable schedule), irregular (nonsupply occurs at unknown intervals, generally for short periods, i.e., no more than a few days), and unreliable (uncertain supply time and long unknown periods of nonsupply) (Galaitsi *et al.* 2016). Past studies suggest that WDS operations in Indian cities fall in irregular and unreliable intermittency (Asian Development Bank & Ministry of Urban Development Government of India 2007; Burt *et al.* 2018). To meet water demands under such erratic WDS operations, consumers generally practice flexible consumption behaviors (Figure 1) that involve withdrawing the available water during shorter supply periods and storing it in intermediate storage facilities (Mohan & Abhijith 2020; Randeniya *et al.* 2022).

From a detailed inspection, consumer behavior in IWS (irregular and unreliable) systems is distinguishable into Types A and B. Suppose the expected static pressures at the consumer (ferrule) points are less, or the supply is essentially uncertain, then the consumers tend to practice Type A behavior (Figure 1(a)) – store water within underground (sumps) storage tanks and later pump it into overhead (rooftop) tanks for use. On the contrary, if the static pressures at the consumer points are high enough to deliver the water to the rooftop tanks directly, the consumers practice Type B behavior (Figure 1(b)) – avoid sump storage and carry out successive pumping. Intriguingly, against the common assumption that the demand satisfaction in residential areas is minimal under IWS, practicing either of these two types of behavior is anticipated to allow the consumers to cope with the noncontinuous water availability and meet their demands more suitably (Andey & Kelkar 2009; Randeniya *et al.* 2022).

Several attempts toward modeling the hydraulics within IWS systems can be seen in the literature (Liou & Hunt 1996; Bourdarias *et al.* 2008; Marchis *et al.* 2010; Mohapatra *et al.* 2014; Leib *et al.* 2016; Campisano *et al.* 2018; Taylor *et al.* 2019; Mohan & Abhijith 2020; Randeniya *et al.* 2022; Suribabu *et al.* 2022; Weston *et al.* 2022; Sinha *et al.* 2023). However, consumer behavior in IWS systems has not been given the needed importance barring a few studies (Mohan & Abhijith 2020; Randeniya *et al.* 2022; Suribabu *et al.* 2022).

Mohan & Abhijith (2020) represented the outflows (water withdrawal) at the consumer points of *highly* (unreliable) intermittent WDS as uncontrolled orifice-based demand. They employed the uncontrolled-type head–discharge relationship suggested by Reddy & Elango (1989) and a noniterative time-stepping method using backward differencing to analyze the hydraulic characteristics of a draining IWS system. However, Mohan & Abhijith (2020) did not emphasize the water storage constraints (volume of sumps and/or rooftop tanks) and inadvertently associated the consumer demands with the leakage losses. Furthermore, their model also showed limitations in conserving mass and momentum and accurately estimating the draining time (Meyer & Ahadzadeh 2021).

Recently, Randeniya *et al.* (2022) and Suribabu *et al.* (2022) proposed distinct yet similar, straightforward methodologies to simulate the hydraulic behavior of the domestic storage tanks and analyzed the performance of IWS systems by applying the modeling environment of EPANET 2.2 (Rossman *et al.* 2020). On the basis of a comprehensive social survey conducted in Kakkapalliya, Sri Lanka, Randeniya *et al.* (2022) reported the attributes of domestic storage tanks and consumer experience in a typical IWS system. After attaching hydraulically equivalent domestic tanks to demand nodes, they constructed the hydraulic model of an intermittently operated WDS in EPANET 2.2 and analyzed the equity and performance under varying IWS operation schedules. However, the approach by Randeniya *et al.* (2022) was limited only to Type B consumer behavior. Such behavior by the IWS system consumers necessitates the availability of optimal or near-to-optimal pressure heads at the consumer points of the distribution network on a sustainable basis, which is rare in many real-world IWS systems (Cook *et al.* 2016; Strijdom *et al.* 2017; Ghorpade *et al.* 2021; Satpathy & Jha 2022). On the contrary, although the approach by Suribabu *et al.* (2022) was related to Type A consumer behavior, their study was limited to understanding the direct effects of filling the underground sump tanks (via uncontrolled withdrawal of water) on emptying the source tank. The impact of actual water consumption (meeting of demands) and subsequent emptying and filling (via pumping) of rooftop tanks on governing the water withdrawal at the consumer points and the overall behavior of a WDS during the IWS practice was not examined.

This article presents a pressure-dependent volume-driven approach and a novel simulation tool (named EPyT-IWS) to overcome the aforementioned limitations of the existing state of the art. The proposed tool allows us to simulate consumer

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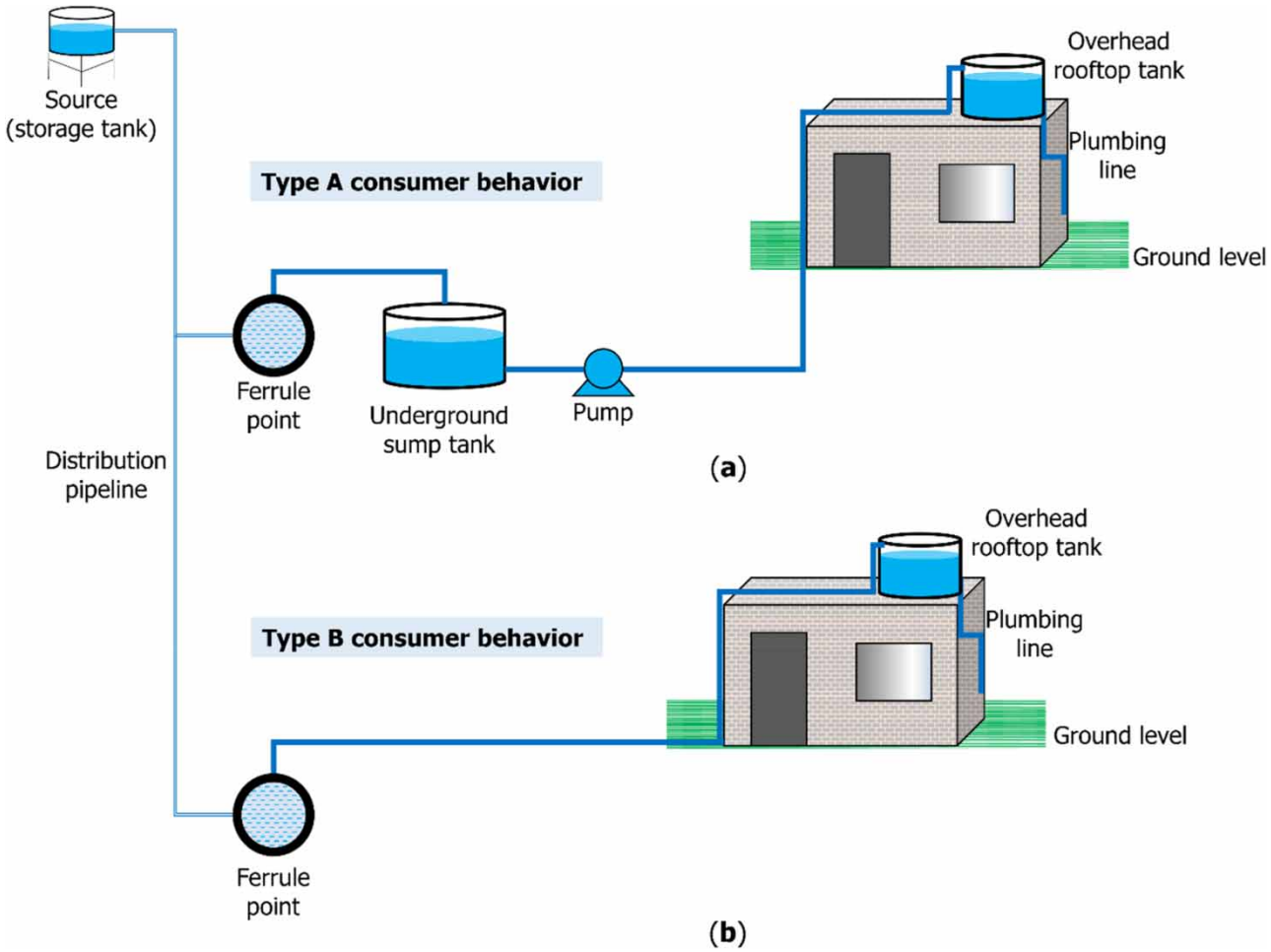

Figure 1 | Schematic representation of (a) Type A and (b) Type B consumer behavior.

behavior in IWS systems realistically and comprehend whether this enables the consumers to cope with the restricted water availability. The suitability of EPyT-IWS for conducting the extended-period analysis was demonstrated by applying it to a typical WDS, delivering water to 2,650 consumers (530 households) initially proposed by Suribabu *et al.* (2022), with the essential characteristics of a representative IWS system. The results corroborate the ability of the proposed tool to mimic the domestic consumer response to intermittency closest to reality and to perceive the knowledge of how far away the WDS is from an equitable operation.

DEVELOPMENT AND FUNCTIONING OF EPYT-IWS

Conceptual overview

EPyT-IWS is designed as an open-source Python-based extension of EPANET 2.2, employing the EPANET-Python toolkit developed initially by the KIOS Research and Innovation Center of Excellence, University of Cyprus. It integrates the modeling environment of EPANET with an independent hydraulic solver. At each time step, the EPANET solver is applied to solve the part of the hydraulic model initialized as input data files for EPANET (in .inp format), and the independent hydraulic solver is employed to solve the remaining part, autogenerated by the EPyT-IWS algorithm. The functioning of EPyT-IWS is conceptualized in Figure 2.

The foremost task in simulating irregular and unreliable intermittent operation of WDS using EPyT-IWS is conceptualizing and developing the hydraulic model. EPyT-IWS defines WDS as graphs, with the links representing the pipes and the nodes representing connections among pipes, consumers, hydraulic control elements (pumps, valves, regulators), and water sources

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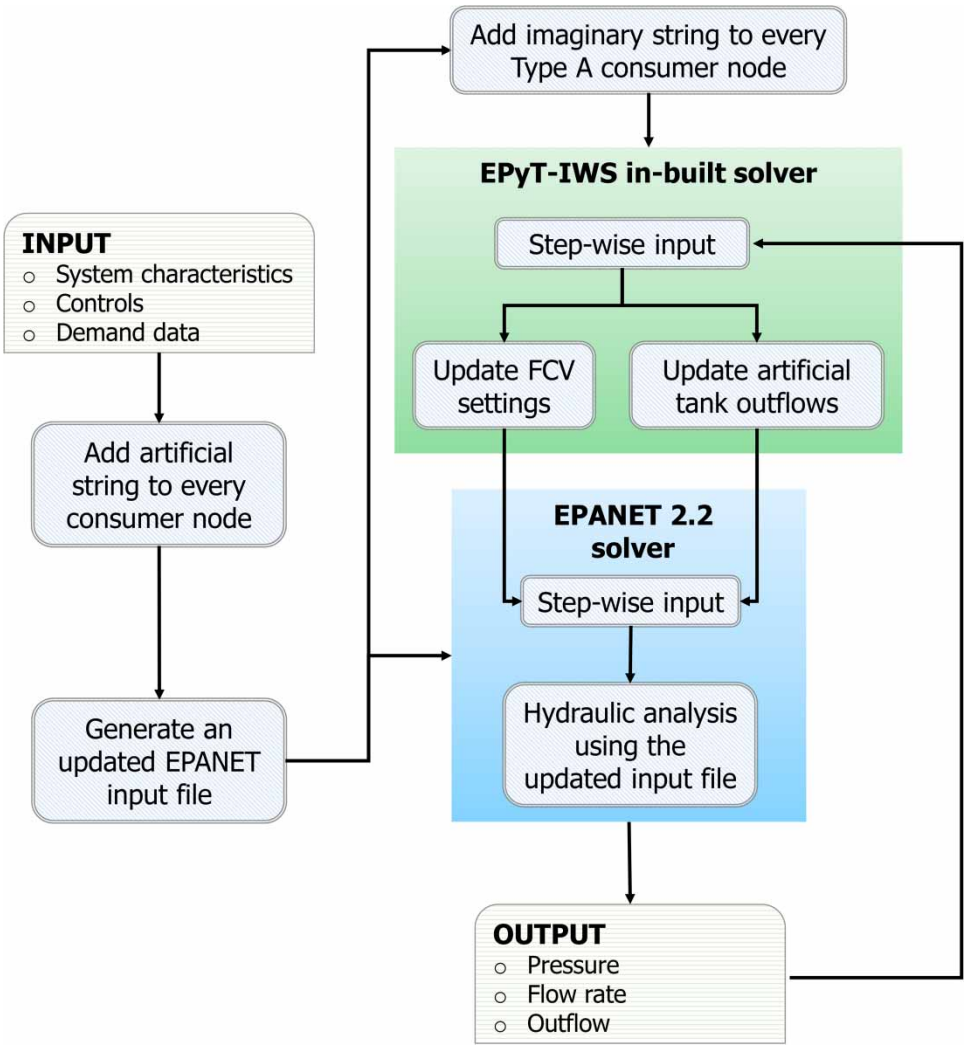

Figure 2 | Conceptual representation of EPyT-IWS functioning.

(reservoirs and tanks). The base demand value is kept at zero at every consumer node, and an artificial string is connected (Figure 3) to simulate consumer behavior associated with IWS operation. For consumer behavior Type A, an imaginary string gets additionally connected to every artificial string (Figure 3(a)), but for Type B, such an imaginary string would be redundant (Figure 3(b)). The imaginary string is autogenerated by the EPyT-IWS algorithm, and its components are not defined in the EPANET input data file.

The artificial string comprises a tank, three nodes, three pipes, and a flow control valve (FCV), and the imaginary string constitutes a tank, a node, and a connecting pipe. Specific nodes and pipes of the artificial strings are dummies. Dummy nodes have zero base demand and the same elevation as the consumer node, except artificial node-3, which has the same elevation as the household. All dummy pipes have a negligible length. The constituents of the artificial and imaginary strings for Type A and Type B consumer behavior are detailed in Table 1.

Simulating withdrawal at consumer nodes

In typical (irregular and unreliable) IWS systems, the house connection constitutes a minimum of three 90° Elbows, a stop valve, and a pipe that discharges freely into a storage tank, all connected in series (Suribabu *et al.* 2022). The minor head losses within the house connection would be significant, and the minor loss coefficient (k_L) would add up to ~5.90

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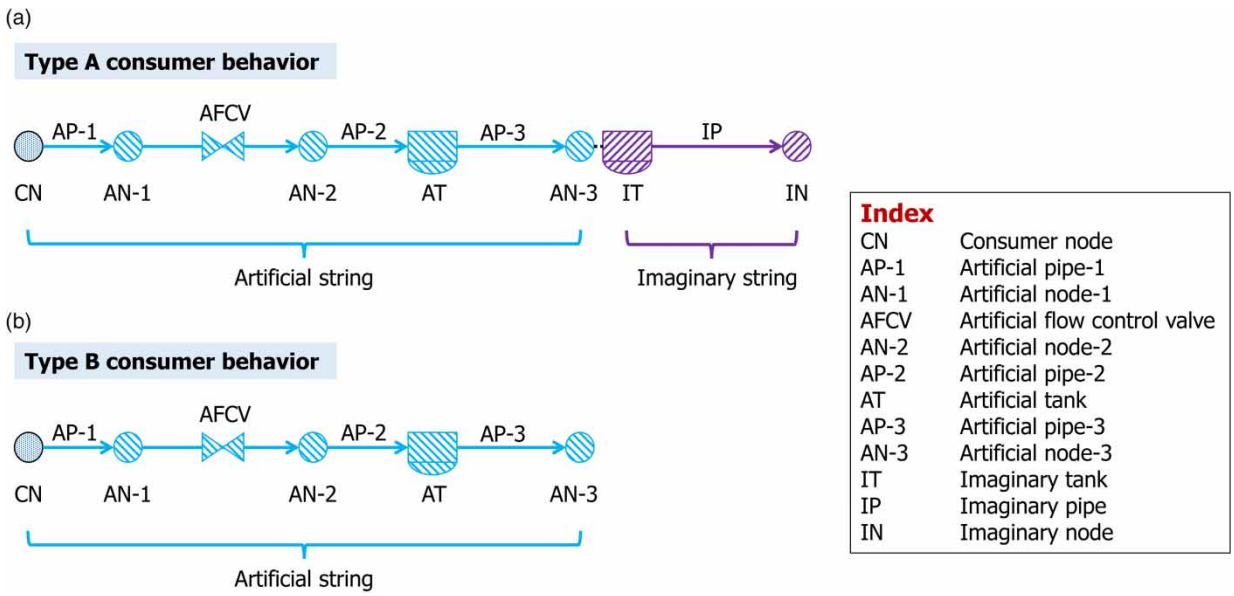

Figure 3 | Artificial and imaginary strings connected to a consumer node with (a) Type A and (b) Type B behavior.

Table 1 | Components of artificial and imaginary string connected to every consumer node in the hydraulic model

	Characteristics		Remarks	
Component type	**Type A**	**Type B**	**Type A**	**Type B**
Artificial tank	Underground sump	Rooftop tank		
Artificial node 1	Dummy node			
Artificial node 2				
Artificial node 3	Dummy node	Effective consumer node		
Artificial pipe 1	Dummy pipe		Connects consumer node and artificial node 1	
Artificial pipe 2	House supply pipe		Connects artificial node 2 and artificial tank	
Artificial pipe 3	Dummy pipe	House connection pipe	Connects artificial tank and artificial node 3	
Artificial FCV	Controls flow rate in house connection pipe		Connects artificial node 1 and artificial node 2	
Imaginary tank	Rooftop tank	–		
Imaginary node	Effective consumer node	–		
Imaginary pipe	House connection pipe	–	Connects imaginary tank and imaginary node	–

(Bhave 1991). Therefore, the flow rate through the house connection pipe at every consumer node would be directed by the static pressure head at the ferrule point and the friction losses within the house connection arrangement.

An approximate technique is applied in EPyT-IWS to determine the flow rate (Figure 4). The pseudo-code for the same is provided in the Supplementary Information. This technique assumes that free discharge occurs at the underground sump tanks (Type A consumers) or the overhead rooftop tanks (Type B consumers). For every consumer node, the major ($f_{L,j}^{\text{major}}$) and minor head loss ($f_{L,j}^{\text{minor}}$) functions are estimated using Equations (1) and (2), respectively.

$$f_{L,j}^{\text{major}} = \frac{10.67 \times L}{C_{\text{HW}}^{1.852} \times D^{4.87}} \tag{1}$$

$$f_{L,j}^{\text{minor}} = \frac{8 \times k_L}{g \times \pi^2 \times D^4} \tag{2}$$

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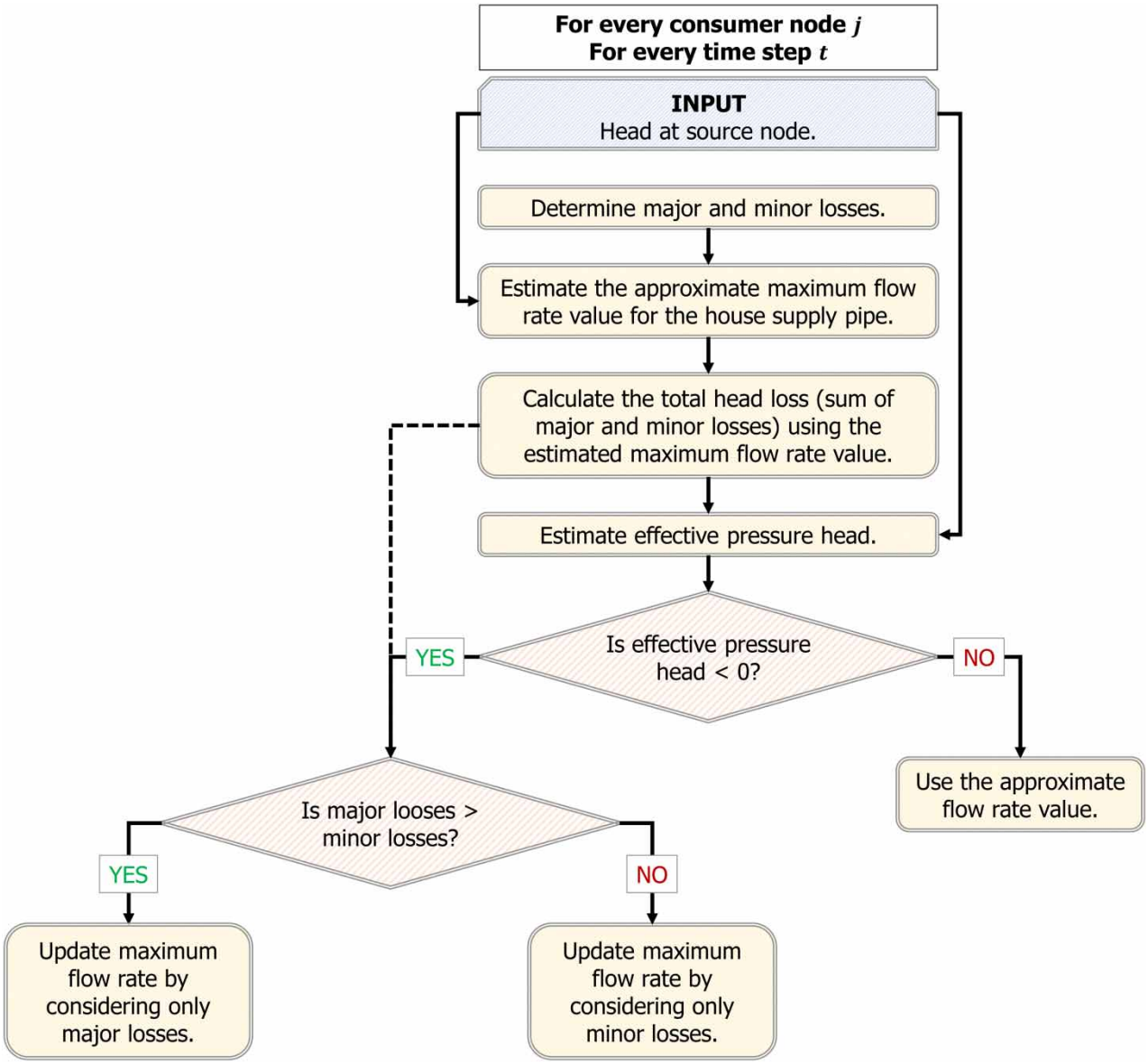

Figure 4 | Conceptual representation of approximate technique to determine house connection pipe flow rate.

where j = node index, L = length of the house connection pipe (m), C_{HW} = Hazen–Williams roughness coefficient value, D = equivalent pipe diameter (m), and g = acceleration due to gravity (m.s^{-2}). The unit of $f_{L,j}^{major}$ and $f_{L,j}^{minor}$ is approximated as s^2.m^{-5}.

Assuming N is the number of house connections aggregated at any consumer point, and the value of D can be estimated using Equation (3):

$$D = d \times N^{0.38} \tag{3}$$

where d = diameter of the individual house connection pipe (m).

At each time step (t), the maximum feasible head loss at every consumer node will be the difference between the piezometric head (H_s^t in m) available at the source node (denoted as s) delivering water to the consumer node j and the datum head available at the consumer node. This is defined using Equation (4):

$$\Delta h_j^t = H_s^t - (z_j + \Delta z_j) \tag{4}$$

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where Δh_j^t = maximum available pressure head at consumer node j for delivering water to the storage tank (m); z_j = elevation of consumer node j (m); and Δz_j = elevation difference between consumer node j and the storage tank (m). The Δh_j^t value for every node j at any time t is calculated based on the assumption that there is no loss in the head between the source node and the consumer node, and the underground sump tank or the overhead rooftop tank level is the minimum.

Based on the maximum feasible head loss (Δh_j^t) at every consumer node j at time t, the maximum flow rate possible for house connection pipe under the free discharge condition can be estimated using Equation (5).

$$\tilde{Q}_{\max,j}^t = \left(\frac{\Delta h_j^t}{f_{L,j}^{\text{major}} + f_{L,j}^{\text{minor}}} \right)^{0.5} \tag{5}$$

where $\tilde{Q}_{\max,j}^t$ = approximate maximum flow rate value for house connection pipe ($m^3.s^{-1}$).

Based on the estimated $\tilde{Q}_{\max,j}^t$ value, the major, minor, and total head losses can be estimated using Equations (6)–(8).

$$h_{L,j}^{\text{major}} = f_{L,j}^{\text{major}} \times \tilde{Q}_{\max,j}^t 1.852 \tag{6}$$

$$h_{L,j}^{\text{minor}} = f_{L,j}^{\text{minor}} \times \tilde{Q}_{\max,j}^t 2 \tag{7}$$

$$h_{L,j} = h_{L,j}^{\text{major}} + h_{L,j}^{\text{minor}} \tag{8}$$

where $h_{L,j}^{\text{major}}$, $h_{L,j}^{\text{minor}}$, and $h_{L,j}$ are the major, minor, and total head loss within the house connection pipe (m), respectively.

Based on $h_{L,j}$ value estimated, the effective pressure head ($\widehat{\Delta h_j^t}$ in m) can be estimated with Equation (9) to verify whether the Δh_j^t value is high enough to permit a flow rate of $\tilde{Q}_{\max,j}^t$ within the house connection pipe.

$$\widehat{\Delta h_j^t} = \Delta h_j^t - h_{L,j} \tag{9}$$

If $\widehat{\Delta h_j^t}$ comes out as a positive value, the withdrawal at node j (q_j^t in $m^3.s^{-1}$) can be as high as the estimated $\tilde{Q}_{\max,j}^t$ value. If not, the q_j^t value satisfying the pressure head constraints must be independently determined. Noniteratively determining this value for each node j at every time t is practically difficult. Therefore, we have adopted an approximate technique (Equation (10)) that considers either the maximum major or minor losses (estimated based on the maximum flow rate value) to speculate a practically appropriate flow rate value that avoids the occurrence of negative pressures at the consumer points. The approximate technique is adopted only to improve the computational efficiency of EPyT-IWS.

$$q_j^t = \begin{cases} \left(\dfrac{(\Delta h_j^t - h_{L,j}^{\text{major}}) \times C_{\text{HW}}^{1.852} \times D^{4.87}}{10.67 \times L} \right)^{1/1.852}, & \text{if } h_{L,j}^{\text{major}} \geq h_{L,j}^{\text{minor}} \\ \left(\dfrac{(\Delta h_j^t - h_{L,j}^{\text{minor}}) \times g \times \pi^2 \times D^4}{8 \times k_L} \right)^{0.5}, & \text{if } h_{L,j}^{\text{major}} < h_{L,j}^{\text{minor}} \end{cases} \tag{10}$$

At each time step, the EPyT-IWS algorithm computes the q_j^t value for every consumer node j and updates the flow setting of the artificial FCV within the artificial string. The flow rate through the artificial pipe 1 and artificial pipe 2 will equal the FCV setting if the artificial tank level is less than its depth. Else, the flow rates would be zero.

Simulating storage tank(s) dynamics

The q_j^t value would govern the inflow to the artificial tank connected to every consumer node. For Type B consumer behavior, the outflow from the artificial tank would be administered entirely by the time-series demand value of artificial node 3. However, for Type A consumer behavior, the imaginary string components need to be accounted for while estimating the artificial tank outflow.

Typically, under Type A consumer behavior in IWS systems, pumps are operated to fill the rooftop tanks and to meet water demands. EPyT-IWS avoids physically adding the pumps to the hydraulic model but indirectly simulates the pump operation

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using the in-built hydraulic solver. The hydraulic solver takes an adequate pumping rate ($q_{\min,j}^{\text{pump}}$) for every consumer node to simulate the imaginary tank filling and artificial tank emptying. The EPyT-IWS algorithm assumes that the demands at consumer nodes are met at any time step t if the water level within the imaginary tank is above a certain value (minimum required water depth, d_{minimum} in m). Similarly, the algorithm assumes no pumping occurs until the water level within the imaginary tank is below a specific depth (threshold depth for pump activation, $d_{\text{threshold}}$ in m). The algorithm considers that even when the imaginary tank level is below the $d_{\text{threshold}}$ value for pump activation, the pump cannot operate until the water level within the artificial tank is above a critical value (critical water depth, d_{critical} in m). Once the pump switches on, the artificial tank is emptied, and the imaginary tank filling would continue until either of the two conditions occurs – the artificial tank level dropping below d_{critical} value or the imaginary tank reaching full.

Extended-period analysis of WDS operation

For performing the extended-period hydraulic simulation using EPyT-IWS, appropriate hydraulic time step (Δt in s) and simulation time duration (T_{sim} in days) values must be specified as input. Special care needs to be given while specifying Δt and T_{sim} values. Typically, 1,800 or 3,600 s is selected as the Δt value for hydraulic analysis using EPANET 2.2. Nonetheless, the Δt value of this scale might make it numerically challenging and time consuming for the EPANET solver to derive numerical solutions due to the necessity of simulating many tanks additionally incorporated into the hydraulic model by the EPyT-IWS algorithm (Randeniya *et al.* 2022). For this reason, a relatively shorter time step in the scale of 60 s is deemed adequate for analysis with EPyT-IWS. Concerning T_{sim}, a longer duration (in the range of weeks or preferably months) should be selected to fully overcome the effects of the initial conditions (levels of the underground sump tank and/or overhead rooftop tank) on the EPyT-IWS outcomes. Besides Δt and T_{sim}, the initial state of the components of the artificial strings and imaginary strings attached to the system nodes of the WDS needs to be indicated. By default, EPyT-IWS takes the initial water level within the artificial and imaginary tanks as 0 and 0.5 m, respectively. Also, the default $q_{\min,j}^{\text{pump}}$ value at every consumer node j is selected as five times its base demand value. The values for d_{minimum}, $d_{\text{threshold}}$, and d_{critical} are set as 0.1, 0.2, and 0.5 m by default.

APPLICATION OF EPYT-IWS

Test network

The rural WDS (Figure 5), initially proposed by Suribabu *et al.* (2022), was slightly modified by introducing a pumping arrangement and was selected as the test network to demonstrate the applicability of EPyT-IWS. The rural WDS serves 530 houses distributed along the pipes. However, the house supply connections were aggregated to nodes for modeling purposes. Each consumer node was presumed to consist of half the number of house supply connections along the pipes connecting them. Each house was assumed to have five inhabitants, and the total number of consumers for the rural WDS was worked out as 2,650. The per capita water consumption was selected as 55 litres per capita per day (LPCD) (CPHEEO 1999).

The elements of the test network include a reservoir (a proxy for a water treatment plant), a pump, an elevated storage tank, 22 nodes, and 22 pipes. The reservoir was assumed to have an elevation of 95 m and was located far from the storage tank. The pumping operation to fill the storage tank was assumed to occur at a fixed pumping rate of 150 litres per minute (LPM). The storage tank has an elevation of 107.5 m, and all consumer nodes have 100 m elevation values. The Hazen–Williams roughness coefficient value for all 22 pipes was assumed as 130. The storage tank outlet was always assumed to be open, causing the tank filling due to pump operation and its draining due to withdrawal at consumer points to happen concurrently. The details regarding the number of house supply connections corresponding to each consumer node, its average daily demand, and the length and diameter of pipes are specified in Table 2.

Test cases for analysis

Different test cases (Table 3) were selected by varying the pump operating schedule to simulate the consumer behavior in IWS systems. For all test cases, the time window was kept at 120 days to evaluate the ability of consumers to cope with the restricted and uncertain water supply practices over the long run of an IWS system. For 9 (out of 11) cases, except Cases 5a and 7a (Table 3), the consumer behavior in every system node was assumed as Type A. Cases 1–3 correspond to consistent operation of pumps for only 1 (from 6 AM to 7 AM), 3 (from 6 AM to 9 AM), and 6 h (from 6 AM to 12 PM) every day, respectively. Cases 4–7 correspond to unreliable intermittency. For all these four cases, the water supply practice for the first 60 (out of 120) days was kept alike: 3 h pumping (from 6 AM to 9 AM) for the first 30 days; no pumping and no water supply during 31–45 days; and restarting of pumping on the 46th day and only 1 h pumping (from 6 AM to 7 AM) for

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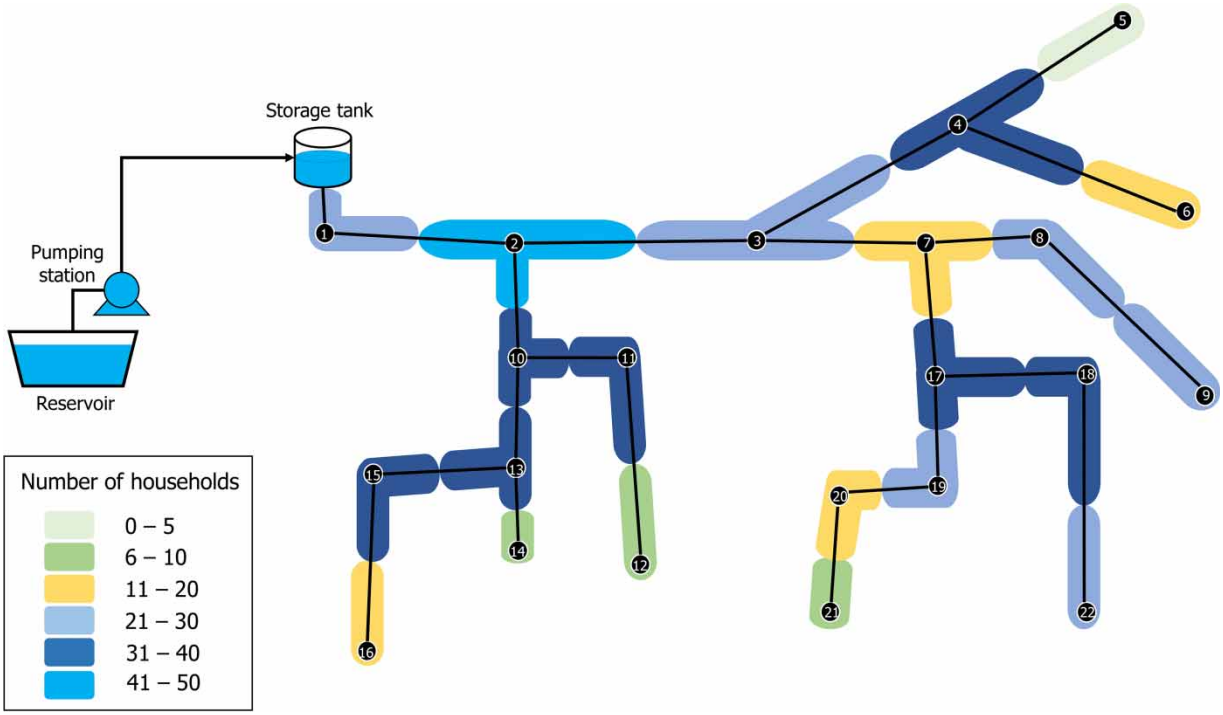

Figure 5 | Schematic of the test network.

the next 15 days. For the subsequent 60 days, 3 h pumping was adopted for Cases 4–6 by varying the clock time of pumping (Table 3). For Case 7, 13 h pumping (from 11 AM to 12 AM) was adopted. Cases 5a and 7a correspond to the case with the consumers of the most upstream node (Node 1) adopting Type B consumer behavior, and the remaining consumers aggregated in the rest of the nodes continuing with Type A behavior. The diurnal variations in the household water consumption at every node under all the cases analyzed were represented using a demand pattern (Table 4) that was evolved from scrutinizing the time-series water demand data generated by Jethoo & Poonia (2011) for an Indian city (Jaipur) based on the socioeconomic data. The consequence of alternate water sources such as groundwater, tankers, bottled water, and community water filters was entirely ignored for modeling purposes.

Performance metrics

Two performance metrics were used to evaluate the EPyT-IWS outcomes and to compare the ability of the considered IWS operations and the water consumption behavior to meet the water demands. The first metric is defined as the demand satisfaction ratio (DSR), which symbolizes the capacity to meet the water demands of the household consumers in any node over a specific time. DSR is specified using an index called demand deficit, defined using Equation (11).

$$d^t_{j,\text{deficit}} = \frac{d^t_{j*,\text{actual}} - d^t_{j*,\text{satisfied}}}{d^t_{j*,\text{actual}}} \times 100 \tag{11}$$

where $d^t_{j,\text{deficit}}$ = demand deficit in consumer node j at time t (%); $d^t_{j*,\text{actual}}$ = actual water demand in node j^* at time t (LPM); and $d^t_{j*,\text{satisfied}}$ = water demand satisfied in node j^* at time t (LPM). For every consumer node j, the node j^* denotes its equivalent imaginary node or artificial node 3 under Type A and Type B consumer behaviors, respectively.

The first performance metric (DSR) can be estimated using Equation (12) based on $d^t_{j,\text{deficit}}$ value.

$$\text{DSR}_{j,T} = \frac{\sum_{t=t_0}^{T} (100 - d^t_{j,\text{deficit}})}{\sum_{t=t_0}^{T} 100} \tag{12}$$

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Table 2 | Node and pipe characteristics of the test network considered

Node characteristics			Pipe characteristics			
			Pipe connecting			
Node ID	Number of house supply connections	Base demand (LPM)	Start node ID	End node ID	Length (m)	Diameter (mm)
1	25	4.775	Tank	1	105	200
2	47	8.977	1	2	300	150
3	29	5.539	2	3	140	150
4	33	6.303	3	4	100	75
5	5	0.955	4	5	60	50
6	18	3.438	3	7	40	100
7	15	2.865	4	6	200	50
8	30	5.730	7	8	60	75
9	25	4.775	8	9	300	20
10	38	7.258	7	17	180	100
11	31	5.921	2	10	72	100
12	8	1.528	10	11	90	75
13	32	6.112	11	12	270	50
14	6	1.146	10	13	96	75
15	33	6.303	13	14	90	50
16	15	2.865	13	15	60	75
17	39	7.449	15	16	210	50
18	38	7.258	17	19	28	75
19	23	4.393	19	20	40	75
20	11	2.101	20	21	170	50
21	6	1.146	17	18	210	100
22	23	4.393	18	22	260	75

Table 3 | Test cases considered for hydraulic analysis

	Time (days)				
Case	1–30	31–45	46–60	61–120	Remarks
1		Pumping from 6 AM to 7 AM			Regular pumping schedule over 120 days
2		Pumping from 6 AM to 9 AM			
3		Pumping from 6 AM to 12 PM			
4				Pumping from 11 AM to 2 PM	Varying pumping schedule over 120 days
5				Pumping from 9 PM to 12 PM	
5a	Pumping from 6 AM to 9 AM	No pumping	Pumping from 6 AM to 7 AM		
6				Pumping from 4 PM to 7 PM	
7				Pumping from 11 AM to 12 AM	
7a					

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Table 4 | Water demand pattern selected

Clock time	Multiplier to the base demand
12 AM–6 AM	0.089
6 AM–10 AM	1.933
10 AM–12 PM	2.133
12 PM–6 PM	0.815
6 PM–8 PM	2.578
8 PM–12 AM	0.356

where $\mathrm{DSR}_{j,T}$ = DSR value for consumer node j over the time window T and t_0 = initial time. $\mathrm{DSR}_{j,T} = 1$ signifies that the consumers in node j can meet their entire demand over time T.

The second performance metric is relative withdrawal (RW) and is defined using Equation (13).

$$\mathrm{RW}_{j,T} = \frac{\sum_{t=t_0}^{T} q_j^t}{\sum_{j=1}^{N_J} \sum_{t=t_0}^{T} q_j^t} \tag{13}$$

where $\mathrm{RW}_{j,T}$ = RW value for consumer node j over the time window T and N_J = total number of consumer nodes.

For every consumer node j, a desirable RW value ($\mathrm{RW}_{j,\mathrm{desirable}}$) can be evolved by associating the number of house supply connections aggregated at the node j with the total number of house supply connections served by the WDS. For example, for Node 1, the desirable RW value would be 0.047 (i.e., 25/530). If the $\mathrm{RW}_{j,T}$ value for every consumer node is equal to $\mathrm{RW}_{j,\mathrm{desirable}}$, then the IWS practice adopted can be deemed equitable over time T.

Input parameters for hydraulic simulation

The extended-period hydraulic analysis of the test network under varying cases (Table 3) was conducted using EPyT-IWS by selecting the Δt as 60 s and T_{sim} as 120 days. The default values were selected for initial water depth within the artificial and imaginary tanks and for $q_{\mathrm{min},j}^{\mathrm{pump}}$. The other input parameter values selected are detailed in Table 5. The simulations were run using a desktop computer with Intel® Core™ i7-6700 CPU @ 3.40 GHz processor, 16.0 GB installed memory, and a 64-bit operating system. It may be noted that due to data paucity, the design of the consumer system (dimensions of the tanks, characteristics of the house connection network, domestic pumping characteristics) was arrived at depending on the

Table 5 | Input parameter values used for extended-period hydraulic analysis with EPyT-IWS

Component type	Characteristic		Value
Artificial tank	Underground sump tank	Maximum capacity (m^3)[a]	2
		Maximum depth (m)	2.5
		Elevation difference between ferrule point and tank top (m)	0.5
	Overhead rooftop tank	Maximum capacity (m^3)[a]	1
		Maximum depth (m)	2
		Elevation difference between ferrule point and tank top (m)	7
Artificial pipe	House connection pipe (connecting ferrule point and storage tank)	Diameter (mm)[a]	19.05
		Hazen–Williams roughness coefficient	100
	House supply pipe (within the household plumbing system)	Diameter (mm)[a]	12.69
		Hazen–Williams roughness coefficient	100

[a]Corresponding to a single household.

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restricted field knowledge of the consumer behavior in IWS systems. Several assumptions have also been made to oversimplify the components of the artificial and imaginary strings.

RESULTS AND DISCUSSION

Effects of pumping duration

The outcomes of the hydraulic analysis of the test network under Cases 1–3 using EPyT-IWS shed light on the effects of pumping duration on the dynamics of water level within the storage tanks and subsequent demand satisfaction of the consumers. Six nodes (Nodes 1, 3, 6, 13, 16, and 22) of the test network were selected (Figure 5) to illustrate and explain the results. Node 1 is the upstream node nearest the storage tank, and Nodes 6, 16, and 22 are downstream dead-end nodes. Nodes 3 and 13 can be defined as intermediate nodes. The variations in the underground sump tank level in the six nodes mentioned earlier over the simulation time under Cases 1–3 are portrayed in Figure 6. Likewise, the predicted water level dynamics within the rooftop tanks are depicted in Figure 7. Figure 8 schematically illustrates the variations in the demand deficit values under the three cases over the simulation period. The values of the performance metrics (DSR and RW) for the 22 nodes over 120 days estimated for Cases 1–3 are shown in Figure 9. The variations of the two metrics over the entire simulation period are also detailed in Figure 9.

Although all six nodes were at the same elevation, the filling and emptying dynamics of the underground sump tanks were found to be entirely distinct for them under the three cases, apparently due to the differences in their specific positions relative to the storage tank. Under Case 1, EPyT-IWS predicted that the water level within the underground sump tank in Node 1 would reach a maximum within ~50 min. However, during the same time, the water levels within the sump tank in Node 3 were predicted to be only 0.63 m, and that in Node 6 were only 0.04 m. The water level profiles under Case 1 predicted for Nodes 1 and 3 differed distinctly from that for Nodes 6, 13, 16, and 22 (Figure 6(a)). The tank water level was found to attain a pseudo-equilibrium around the d_{critical} value for these four nodes. This is because the EPyT-IWS algorithm envisages the consumers to start the water pumping as soon as the tank level rises above the d_{critical} limit to overcome the water shortage caused by the emptying of rooftop tanks (Figure 7). Intriguingly, the underground sump tank level in Node 1 was never predicted to reach the d_{critical} value, while the same in Node 3 oscillated between 0.5 and 0.6 m during the pumping hours (Figure 6(a)).

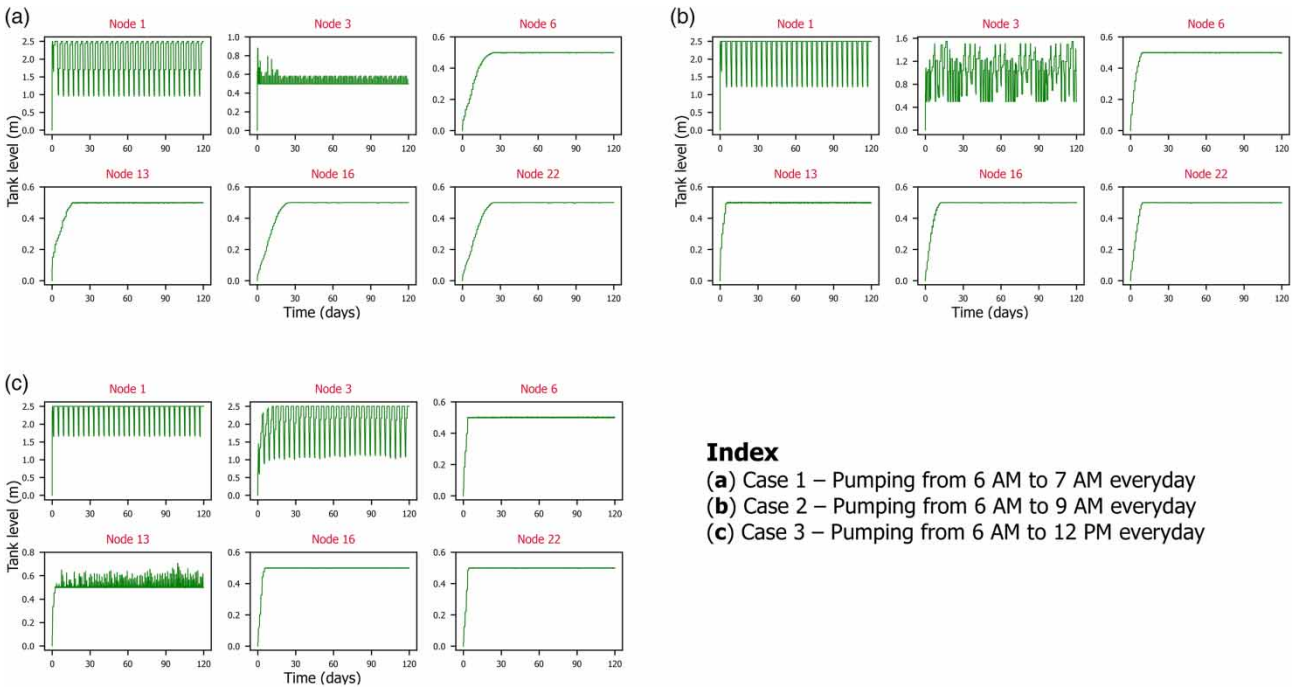

Figure 6 | Underground sump tank level variations in selected nodes predicted over 120 days in (a) Case 1, (b) Case 2, and (c) Case 3.

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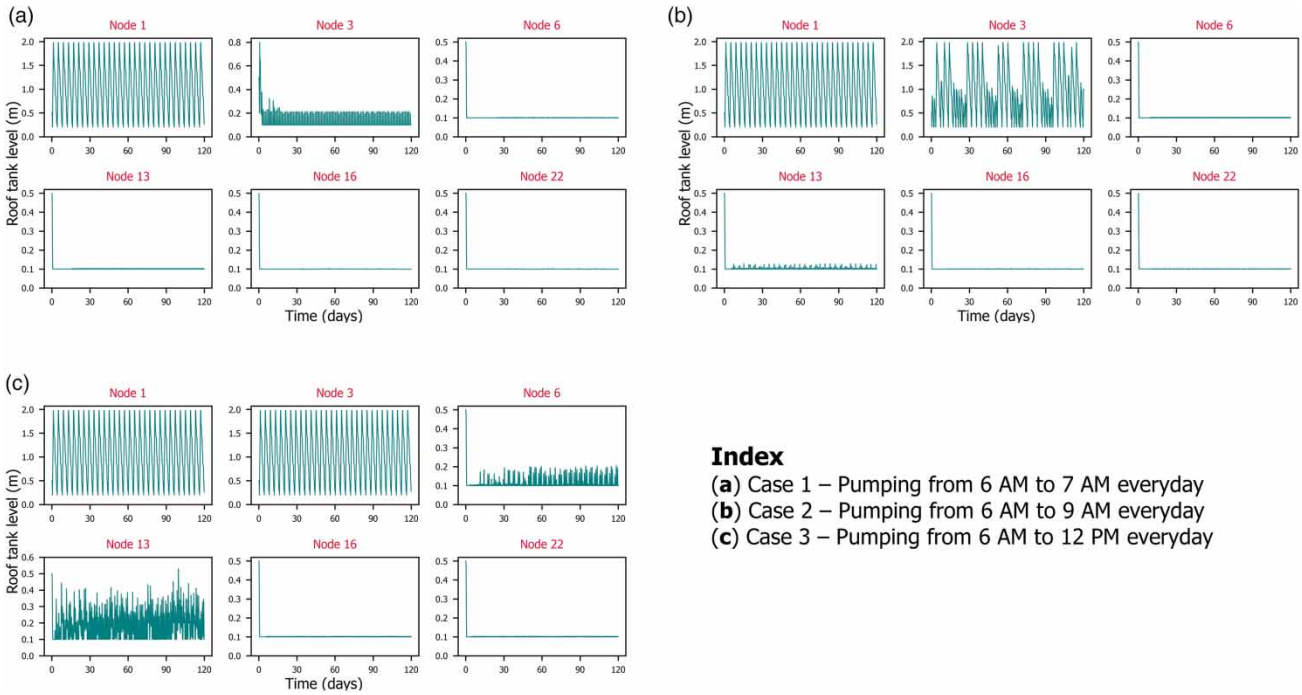

Figure 7 | Rooftop tank level variations in selected nodes predicted over 120 days in (a) Case 1, (b) Case 2, and (c) Case 3.

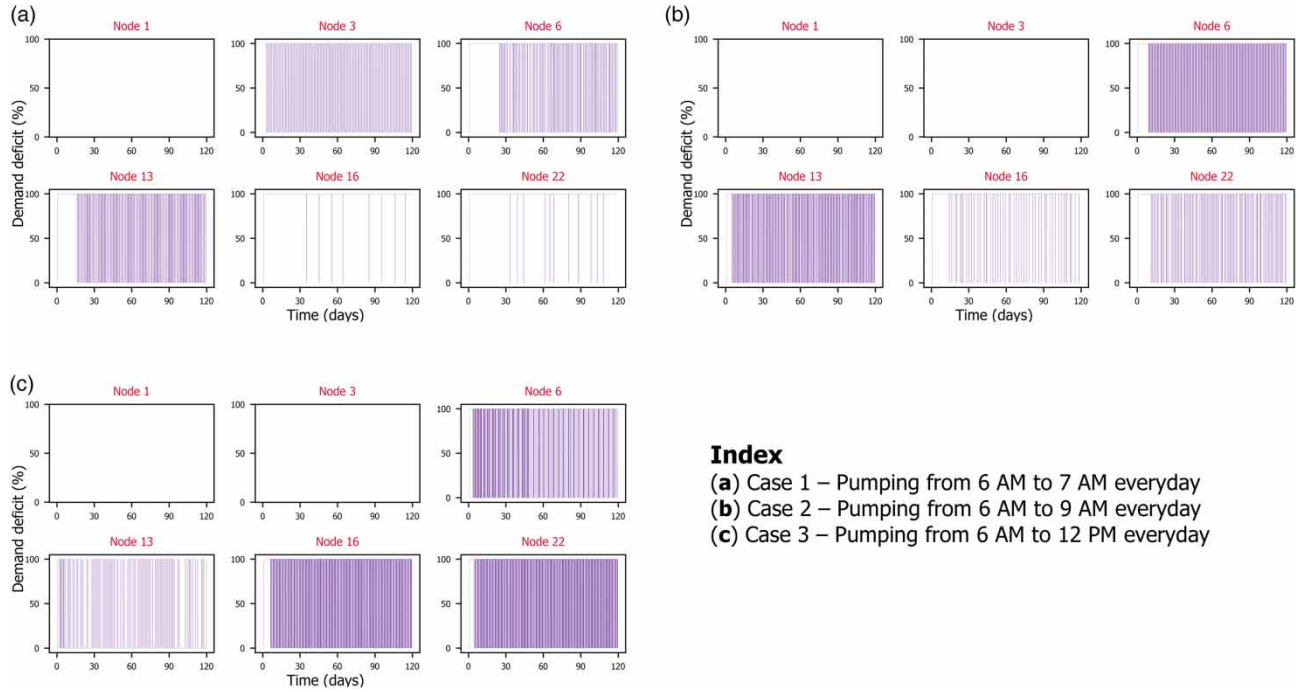

Figure 8 | Demand deficit variations in selected nodes predicted over 120 days in (a) Case 1, (b) Case 2, and (c) Case 3.

By increasing the pumping hours from 1 to 3 h, the minimum depth to which the water level drops within the sump tank in Node 1 increased from 0.96 to 1.22 m, and the highest level reached within Node 3 rose from 0.88 to 1.55 m. An entirely contracting sump tank level profile compared to Case 1 with a mean value of 1.01 m was predicted for Node 3. For the

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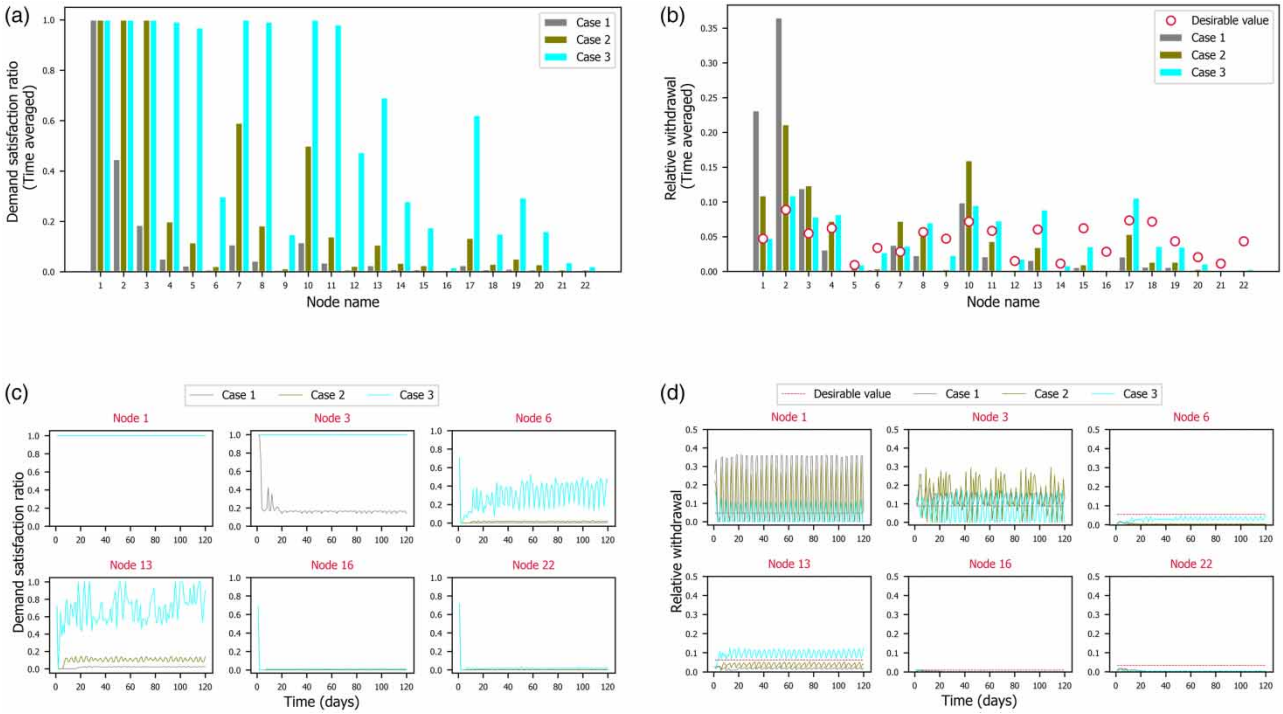

Figure 9 | (a) and (b) DSR and RW values over 120 days corresponding to Cases 1–3 estimated and (c) and (d) their time-series variations predicted in selected nodes.

other four nodes, an escalation of the slopes of the initial rising part of the profile curves was virtually evident (Figure 6(b)). For instance, concerning the sump tank in Node 22, the time required to attain the pseudo-equilibrium decreased from 24.3 days in Case 1 to 9.4 days in Case 2. Similar impacts as Case 2 were also predicted for Case 3 with 6 h regular pumping (Figure 6(c)). The minimum depth to which the water level drops within the sump tank in Node 1 increased further to 1.66 m, and the tank in Node 3 was predicted to get full. From a vague look, the sump tank profiles in Node 3 and Node 13 under Case 3 resembled those predicted in Node 1 and Node 3, respectively, in Case 1 (Figure 6(a) and 6(c)). Concerning Node 22, the time to attain pseudo-equilibrium decreased further to 4.4 days. An increase in the slopes of the early rising portion of the profile curves corresponding to Nodes 6, 13, 16, and 22 was also predicted in Case 3 compared to Case 1.

Concurrently, analyzing the EPyT-IWS predictions of rooftop tank levels (Figure 7) and demand deficit values (Figure 8) under the three different pumping operation cases delivers a perfect picture of the ability of Type A consumer behavior to cope with the IWS practices. As shown in Figure 7, even in Case 1 with only 1 h daily pumping, the household consumers aggregated in Node 1 were predicted to meet their water demands continually without any demand deficit. However, the consumers aggregated in none of the other nodes could meet their demands without deficit under Case 1 (Figure 7(a)). As was predicted for overhead tanks (Figure 6), the rooftop tank water levels in Nodes 6, 13, 16, and 22 under Case 1 attained a pseudo-equilibrium around the d_{minimum} value.

Significant changes in the demand deficit plots became apparent with increased pumping duration (Figure 8). The 3 h pumping practice in Case 2 was found adequate to meet the water demands of all 29 households aggregated in Node 3 incessantly. While the 6 h pumping practice (Case 3) was insufficient to satisfy the water demands of consumers aggregated in every node, the results signified that Type A behavior can facilitate the consumers to lessen the demand deficits by a significant degree (Figure 8(c)). The mean demand deficit values for Nodes 6 and 13 decreased from 99.2 and 97.5% to 70.2 and 30.9%, respectively, after changing the pumping duration from 1 to 6 h. Nonetheless, a similar impact was not predicted for Nodes 16 and 22. The mean deficit values for these nodes over 120 days decreased to 98.3 and 97.9% from a shared value of 99.4%. These disparities, over again, emphasize the importance of the consumer positions relative to the water source in coping with the IWS practices.

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The performance metric values estimated under the three cases also verify the previous inferences. As seen in Figure 9(a), the time-averaged $DSR_{1,120} = 1$ estimated for Node 1 over 120 days under the three cases considered suggests that the household consumers in Node 1 can meet their water demands like that in a continuous WDS by adopting Type A behavior. However, in Case 1, the DSR value estimated for all other nodes, including Node 2, the second-most upstream node, was found to be less than one. The DSR value of 0.446 estimated for Node 2 implied that the 47 households aggregated in this node could fully meet the water demands only ~45% of the time considered. The increment in the pumping hours induced uneven impacts in the DSR values estimated for the 22 nodes. Although a pumping duration of 6 h was found appropriate in fully meeting the demands in five nodes (Nodes 1, 2, 3, 7, and 10) and in meeting water demands for >90% of the time considered in four nodes (Nodes 4, 5, 8, and 11), it was still not enough for consumers in 44 households, mainly aggregated in the three downstream nodes (Nodes 16, 21, and 22), to fulfill demands for >0.05% of the total analyzed time. Furthermore, the variations in the estimated DSR values over the 120 days window (Figure 9(c)) disclosed that even though the typical (Type A) consumer behavior facilitates an improved demand satisfaction, it cannot entirely avoid the unreliability in water availability associated with IWS practices.

The RW values estimated under the three cases revealed more specifics of equity concerning the IWS operation practices. In Case 1, in 5 of 22 nodes (Nodes 1, 2, 3, 7, and 10), the time-averaged RW values over the 120 days were more significant than the corresponding desirable values (Figure 9(b)). This signifies that the consumers of 154 households aggregated in these five nodes draw more water from the WDS during the supply hours than the remaining 376 households. However, surprisingly, although ~36% of the total water supplied over the 120 days (four times the desirable) was predicted to be drawn by the households aggregated in Node 2, the $DSR_{2,120}$ value was estimated to be only 0.446. Interestingly, by drawing only ~21% of the total water supplied over the 120 days in Case 2, the consumers in Node 2 were predicted to be able to fulfill their water demands. This signifies that the RW value only gives a certain sense of the relative distribution of the delivered water among the different nodes and sheds no light on the delivered water volume. As well, looking specifically at Node 1, in Case 1, the 25 aggregated households fully met their demands by drawing ~13% less than that drawn by consumers of Node 2. Although the time-averaged RW values for both Nodes 1 and 2 were more than four times their respective $RW_{j,\text{desirable}}$ values, the consumers aggregated in Node 2 could not fill their underground sump tanks, and the tank level was predicted to oscillate between 0.75 and 0.5 m. Due to this, they could not pump water from the underground sump tank to the rooftop tanks most of the time. Thus, it can be implied that the discontinuity of water availability can impede consumers' capacity to fulfill their demands.

As expected, increasing the pumping hours has positively impacted reducing the disparities between the calculated and expected time-averaged RW values (Figure 9(b)). However, the effects were pronounced only at the nodes closer to the storage tank, and at downstream nodes, particularly in Nodes 16, 21, and 22, the impacts were relatively nominal. For instance, between Cases 1 and 3, the time-averaged RW value calculated for Node 22 only increased from 0.002 to 0.003, while that Node 1 decreased from 0.231 and almost reached the desirable value. The 120-day variations in the RW values also verify the aforementioned inference (Figure 9(d)). Altogether, the EPyT-IWS outcomes corresponding to Cases 1–3 imply the importance of simultaneously examining the performance metrics (DSR and RW) in arriving at the optimal pumping schedule for any IWS system.

Effects of clock time of pumping

The EPyT-IWS outcomes of the test network analysis under Cases 4–7 revealed a picture of consumer response against irregular and unreliable IWS practices in WDS. The results also highlighted the effects of the clock time of pumping on the dynamics of water withdrawal at the consumer points and subsequent demand satisfaction of the consumers. The time-averaged values of DSR and RW over 120 days estimated for Cases 4–7 are depicted in Figure 10. The variations of these two metrics in Nodes 1, 3, 6, 13, 16, and 22 over the entire simulation period are also illustrated in Figure 10. It may be recalled that there is no difference in the total pumping duration among Cases 4–6, and the total period for which the pump operates and supplies water to the storage tank is 285 h over 120 days. The only difference among Cases 4–6 is the clock time of pump operation over the last 60 days of the total time considered (Table 3). Nonetheless, under Case 7, unlike Cases 4–6, the pump operates from 11 AM till 12 AM for an additional 600 h over the last 60 days.

Interestingly, substantial disparities were evident in the time-averaged DSR values estimated for the 22 nodes over 120 days under Cases 4–7 (Figure 10(a)). Although such differences for the time-averaged DSR under Case 7 from the rest were foreseeable, the apparent dissimilarities between the values estimated under Cases 4–6 were unexpected. For Node 1, as anticipated,

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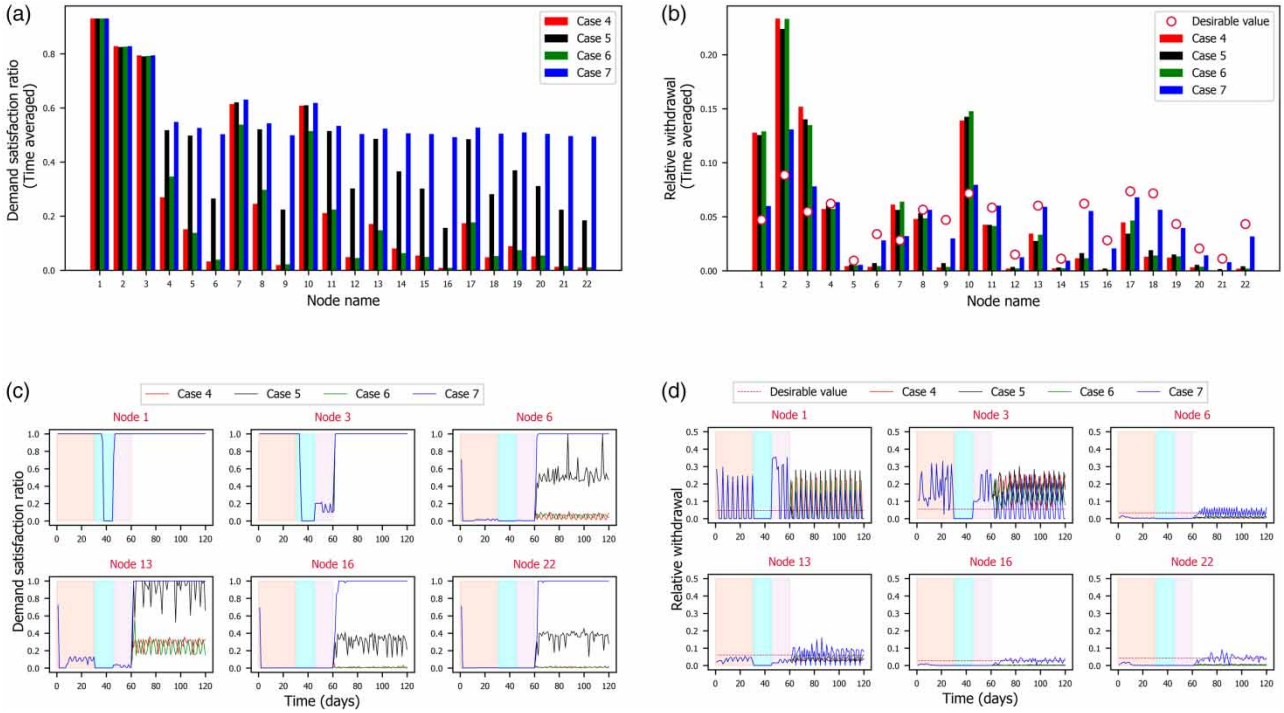

Figure 10 | (a) and (b) DSR and RW values over 120 days corresponding to Cases 4–7 estimated and (c) and (d) their time-series variations predicted in selected nodes. The colored regions correspond to 1–30 days (coral color), 31–45 days (cyan color), and 46–60 days (plum color). Please refer to the online version of this paper to see this figure in colour: http://dx.doi.org/10.2166/hydro.2023.022.

the $DSR_{1,120}$ value estimated (0.931) was the same under all four cases of irregular pumping. However, the value was less than the maximum estimated under Cases 1–3 (Figure 9(a)). This signifies that the amount of water that can be stored within the underground sump tank is insufficient to prevail over the nonavailability of water caused by the shutting down of pumps for 15 days (Figure 10(c)). For Nodes 2 and 3, the time-averaged DSR values were almost equal under Cases 4–6. However, from a closer look, the DSR values under Case 4 were slightly higher than the other two and were equal to that estimated in Case 7 for Nodes 2 and 3 (0.828 and 0.794, respectively). Between the values estimated in Cases 5 and 6, the values in Case 5 were marginally high.

However, for the remaining 13 nodes, the demand satisfaction of consumers in Case 5 was higher than that in Cases 4 and 7. These differences were predicted to be more profound in the downstream nodes. Specifically, looking at Nodes 6, 16, and 22 (Figure 8(a) and 8(c)), operating the pump from 9 PM to 12 AM for the last 60 days (Case 5) induced a substantial rise in the ability of consumers to fulfill their demands compared to operating the pumps from 11 AM to 2 PM (Case 4) and 4 PM to 7 PM (Case 6). The estimated time-averaged DSR values for these three nodes were 0.266, 0.157, and 0.185 in Case 5, while values in Cases 4 and 7 were significantly lower (0.033, 0.009, and 0.010 and 0.039, 0.009, and 0.011, respectively). The reason for the same can be attributed to the diurnal variations in water demand. In Case 5, the pump was operated during the period when the water demands of the consumers were relatively minimal (Table 4). Due to this reason, the consumers aggregated in the upstream nodes (mainly Nodes 1–3), who experience a somewhat uninterrupted demand satisfaction, seldom operate their pumps during the late-night pumping hours. Hence, outflows to the underground sump tanks of households aggregated in the upstream nodes were predicted not to occur for extended periods in Case 5 compared to that in Cases 4 and 7. This led to more volume of delivered water becoming available to withdraw for the underground sump tanks aggregated in the downstream parts of the WDS, eventually enabling them to fill up their rooftop tanks more frequently and fulfill their demands. In Case 7, by increasing the pumping hours from 3 to 13 h for the last 60 days, the time-averaged DSR values for all the nodes of the test network were predicted to attain the maximum value of 1 (Figure 10(c)). This implies that adopting the typical Type A behavior enables the consumers to enjoy uninterrupted demand satisfaction even under the IWS operation of WDS if the water supply is reasonably prolonged and regular (predictable intermittency).

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Although differences in the time-averaged RW values estimated under Cases 4–6 for the 22 nodes are not as significant as those for DSR, noticeable disparities become apparent from a thorough look (Figure 10(b)). Like what we have observed in Cases 1–3, the time-averaged RW values estimated in Cases 4–6 in the upstream nodes were larger, and the time-averaged RW values corresponding to the downstream nodes were significantly lower than the respective desirable values (Figures 9(b) and 10(b)). Although Case 5 can be deemed the most effective pumping schedule among the three irregular pumping cases (Cases 4–6) from the DSR values, it is challenging to make such a clear-cut inference based on the RW values. However, based on the results, it is hard to disregard the growth in the equity of the water supply introduced by the increase in the pumping hours (Figure 10(b) and 10(d)). For Nodes 1, 3, 6, 13, 16, and 22, with desirable values of 0.047, 0.055, 0.034, 0.060, 0.028, and 0.043, respectively, the average RW values corresponding to the final 60 days in Case 5 were 0.071, 0.160, 0.009, 0.029, 0.002, and 0.005. By increasing the water supply duration, the average RW values over the final 60 days improved to 0.040, 0.073, 0.035, 0.066, 0.026, and 0.039, respectively, and became very close to the desirable values (Figure 10(d)). Thus, it can be implied that Type A consumer behavior stimulates a virtually equitable water distribution among consumers under a predictable and regular water supply routine.

Effects of altering consumer behavior

From the previous results (Figures 7 and 8), it became apparent that by adopting Type A consumer behavior, the consumers aggregated in the most upstream node (Node 1) could meet virtually their entire water demands compared to those aggregated in other locations of the test network. Naturally, it becomes appealing to comprehend how this could vary if those consumers switch over to the more economical Type B behavior, avoiding constructing underground sump tanks and operating pumps. In this regard, Cases 5a and 7a were analyzed using EPyT-IWS. The values of DSR and RW thus calculated are illustrated in Figure 11.

By switching from Type A behavior to Type B, the $DSR_{1,120}$ value for the consumers aggregated in Node 1 declined substantially from 0.931 (Figure 10(a)) to 0.007 (Figure 11(a)). This specified the ineptness to move to Type B behavior if the pumping operation continues the irregular and unreliable routine specified as Case 5. The results from EPyT-IWS simulations

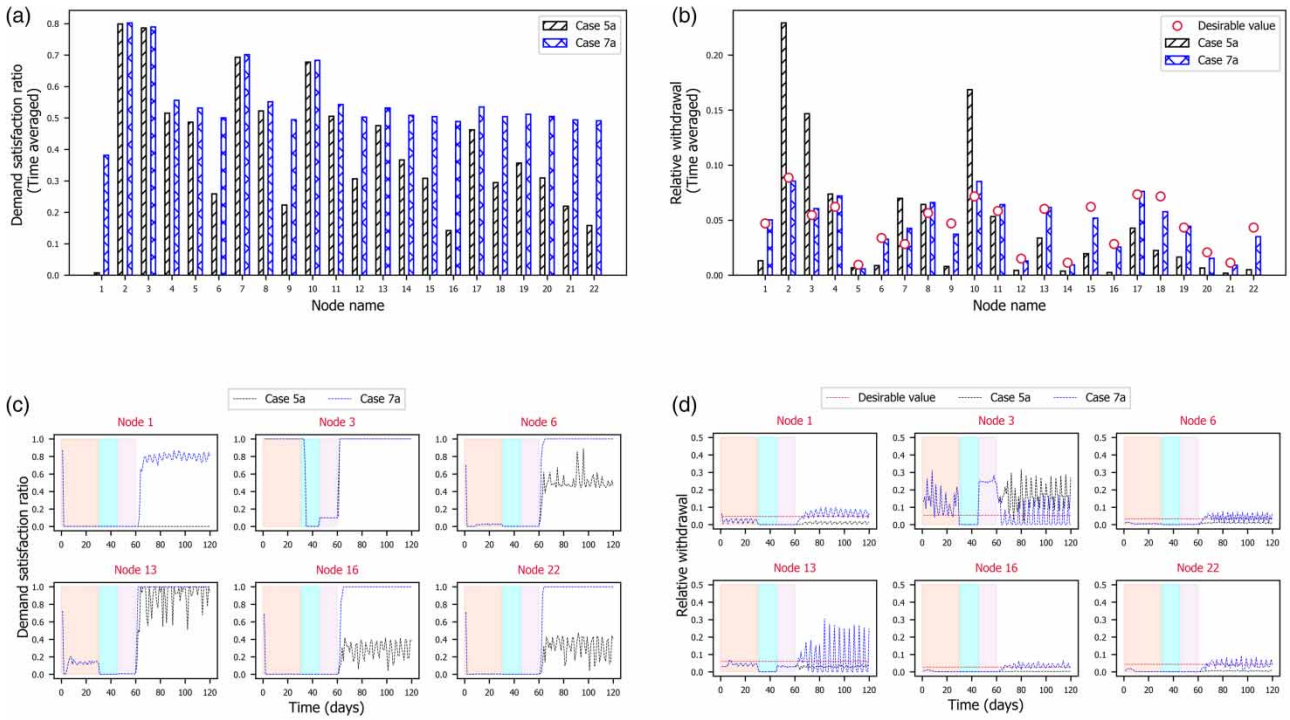

Figure 11 | (a) and (b) DSR and RW values over 120 days corresponding to Cases 5a and 7a estimated and (c) and (d) their time-series variations predicted in selected nodes. The colored regions correspond to 1–30 days (coral color), 31–45 days (cyan color), and 46–60 days (plum color). Please refer to the online version of this paper to see this figure in colour: http://dx.doi.org/10.2166/hydro.2023.022.

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showed that the storage tank, acting as the source to the consumers of the test network, is prevented from becoming full under Case 5a due to the constant outflows to the underground sump tanks during the periods of water availability. Therefore, the static pressure at the ferrule point(s) of Node 1 never reaches the needed value to deliver water directly to the rooftop tanks located at a higher elevation (7 m above the ferrule point). The rooftop tank in Node 1 was predicted to fill for only 35 min (from 0 to 1.05 m) from the start of the simulation, concurrently with the storage tank emptying (from the initial level of 2.85 m). As a result, the consumers aggregated in this node could only satisfy their demands for ~21 h (over 120 days) by withdrawing water from the rooftop tank till it fell to the level corresponding to d_{minimum} (Figure 11(c)). A noticeable decrease in the time-averaged RW values for Node 1 (from 0.126 under Case 5 to 0.013 under Case 5a) also indicated that adopting Type B behavior by the consumers aggregated in Node 1 is not worthwhile under the IWS practice considered (Figures 10(b) and 11(b)).

The inability of households in Node 1 to satisfy demands brought upon by switching over to Type B behavior did not reflect considerably on the remaining 505 households aggregated in the remaining 21 nodes (Figures 10(a) and 11(a)). No noticeable change in the DSR values in Case 5a compared to that in Case 5 was predicted in the downstream nodes (Figures 10(c) and 11(c)). At the same time, a visible increase was evident in two intermediate nodes – Nodes 7 and 10. The time-averaged DSR values in these two nodes increased to 0.693 and 0.677 from 0.621 and 0.610, respectively. Unlike the DSR values, the RW values predicted for the intermediate nodes – specifically Nodes 4, 7, 8, 10, 11, 13, 15, 17, and 18 – displayed an obvious rise between Cases 5 and 5a. The time-averaged RW values for the aforementioned nine nodes increased from 0.063, 0.056, 0.053, 0.143, 0.043, 0.028, 0.016, 0.034, and 0.019 to 0.074, 0.070, 0.064, 0.169, 0.053, 0.034, 0.020, 0.043, and 0.022, respectively (Figures 10(b) and 11(b)).

Switching from Type A to Type B behavior for Node 1 was predicted to be more viable if the IWS practice followed the pumping routine specified by Case 7. The predicted increase in the average DSR values for Node 1 over the last 60 days (from 0 in Case 5a to 0.748 in Case 7a) verifies the above statement (Figure 11(c)). The viability of switching from Type A to Type B was also reflected in the RW values (Figure 11(b) and 11(d)). However, switching to Type B consumer behavior in Node 1 impeded the consumers from achieving complete demand satisfaction (Figure 11). Besides, as predicted (Figure 10), complete demand satisfaction and virtually equitable water distribution can likely be attained by adopting Type A behavior in every node. Considering the aforementioned facts, altering the consumer behavior in Node 1 was deemed unreasonable for the test network under the cases analyzed.

Sensitivity of EPyT-IWS outputs to input parameter values

The sensitivity analysis was performed to ascertain how far the assumptions in developing the hydraulic model impact the EPyT-IWS outcomes. In this direction, a deterministic sensitivity analysis procedure was adopted, and EPyT-IWS simulations were performed by varying the value of one uncertain parameter at a time. The parameters selected include the maximum capacity and maximum depth of the underground sump and overhead rooftop tanks, the diameter values of the house connection pipe (connecting ferrule point and storage tank), and the household pumping rate ($q_{\text{min},j}^{\text{pump}}$). The default values for the maximum capacity of the underground sump tank and overhead rooftop tank were selected as 2 and 1 m^3, respectively. The default values selected for the maximum depth of the underground sump tank, the maximum depth of the overhead rooftop tank, and the diameter of the house connection pipe were 2.5, 2, and 19.05 mm, respectively. For $q_{\text{min},j}^{\text{pump}}$, the default value was five times the base demand value at every node. For the test network, EPyT-IWS simulation was performed by selecting the pumping routine specified in Case 1 using the default values with $\Delta t = 60$ s and $T_{\text{sim}} = 120$ days. The results obtained were reported as the base values.

Once the base values were reported, the input parameters were varied individually. The maximum capacity of the underground sump tank was varied to 1.5 (chosen minimum) by keeping all the other parameters equal to their default value. EPyT-IWS simulation was performed in Case 1 with $\Delta t = 60$ s and $T_{\text{sim}} = 120$ days, and the deviations in the time-averaged DSR and RW values from the base values were established. Likewise, the maximum capacity of the underground sump tank was varied to 2.5 (chosen maximum), and the sensitivity of the value of the maximum capacity of the underground sump tank to the EPyT-IWS predicted time-averaged DSR and RW values were determined. Correspondingly, the maximum capacity of the rooftop tank was varied to 0.5 and 1.5 m^3. From the default values, the maximum depth of the underground sump and overhead rooftop tanks were altered to 2 and 3 m and 1.5 and 2.5 m, respectively. The chosen minimum and maximum values for the diameter of the house connection pipe were selected as 12.69 and 25.40 mm, respectively. From the default value, the $q_{\text{min},j}^{\text{pump}}$ value was varied to three and seven times the base demand value. The results obtained, in terms of time-averaged DSR and RW values, are described in Figure 12.

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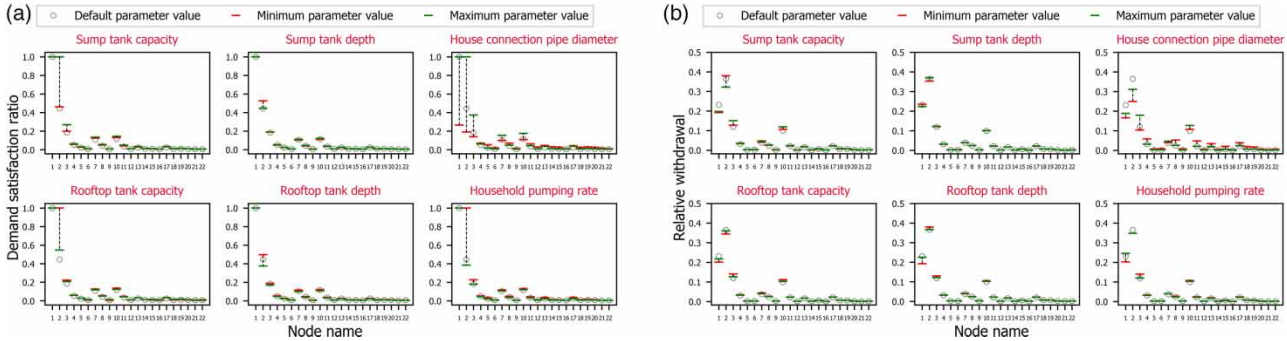

Figure 12 | Sensitivity analysis results obtained. The black dotted line signifies the difference between time-averaged (a) DSR and (b) RW values estimated using minimum and maximum selected values of the parameters.

The results specify that out of the six parameters, the EPyT-IWS outcomes are most sensitive to the diameter value selected for the house connection pipes. Notable differences between the time-averaged DSR and RW over 120 days were evident in all 22 nodes from varying the house connection pipe diameter, contrasting that distinguishable time-averaged DSR and RW value variations were apparent only in the upstream node, primarily Nodes 1 and 2, for the rest five parameters (Figure 12). The high impact of the diameter value of the house connection pipe in controlling the EPyT-IWS outcomes is attributed to the implication of $f_{L,j}^{\text{major}}$ and $f_{L,j}^{\text{minor}}$ values on the pressure-dependent q_j^t value estimation for every consumer node j (based on Equation (5) or (10)). The second-most artificial string component parameter that is most influential to the EPyT-IWS outcomes was the capacity of the underground sump tanks. This value establishes the total volume of water that the consumers can collect during water supply hours for subsequent pumping and usage, influencing the demand satisfaction and the total withdrawal volume at the consumer points. The household pumping rate was found to be the imaginary string component parameter that significantly controls the time-averaged DSR and RW estimations because of its influence on governing the emptying and filling dynamics of the sump tank and rooftop tanks, respectively. The other three parameters, such as sump tank depth, rooftop tank capacity, and rooftop tank depth, were not quite prominent in administering EPyT-IWS outcomes under the IWS operation scenario specified as Case 1.

LIMITATIONS AND SCOPE FOR FUTURE STUDY

The findings corroborate the potential of the EPyT-IWS tool to induce significant contributions toward planning, designing, and operating WDS with IWS characteristics. However, numerous assumptions were made to oversimplify consumer behavior in IWS systems and to develop the hydraulic model, which is yet to be calibrated and verified using real-world data. Leakage is significant in WDS, particularly those operated intermittently (Ghorpade *et al.* 2021). In addition, considerable volumes of water are wasted within the IWS system consumer premises owing to the overflow of domestic storage tanks during water availability or due to these tank emptying preceding the supply course to avail fresh water (Mastaller & Klingel 2018). Nevertheless, these losses were not accounted for in the present study. During IWS operation, air entrapment may occur in pipes, may amplify the frictional resistance to flow (Kumpel & Nelson 2014), and may cause the flow to get transitioned between free surface and pressurized flows (Kerger *et al.* 2012; Leib *et al.* 2016). These complex dynamical features of flows within WDS during IWS operation were overlooked in the EPyT-IWS development. The future directions involve advancing EPyT-IWS toward simulating transient free-surface and pressurized flows and modeling leakage losses.

CONCLUSIONS

This article presented the development, functioning, and application of a pressure-dependent volume-driven hydraulic analysis approach and a novel simulation tool (named EPyT-IWS) for WDS toward accounting for characteristic consumer behavior in understanding the performance of IWS systems more realistically. The following specific inferences were drawn after interpreting the EPyT-IWS simulation outcomes obtained by applying it to a representative IWS system:

- Type A consumer behavior (storing water within underground sump tanks and later pumping it into overhead rooftop tanks for use) enhances the capability of consumers of IWS systems to enjoy improved demand satisfaction even under a limited water supply.

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- The positioning of consumers relative to the source tank is an essential factor in controlling their ability to withdraw water from WDS to their respective storage tanks.
- Barring the water supply duration, the clock time of water availability in IWS systems can significantly influence consumers' ability to fill their storage tanks and meet water demands.
- The sensitivity analysis results established the inevitability of careful selection of the parameter values concerning the additional artificial and imaginary string connected to every node in the hydraulic model on the reliability of EPyT-IWS predictions.

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AUTHOR CONTRIBUTIONS

All authors contributed to the study conception and design. Model development and analysis were performed by G. R. A. The first draft of the manuscript was written by G. R. A., and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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